The Effects of Attention Bias Modification on Aggression in Justice-Impacted Young Adults

Boomer Rawlings

Southwestern College

Psychology 255-501

Professor Contreras

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Introduction

Aggression and poor self-regulation are major concerns in justice-involved populations, especially in young people who are moving through detention, incarceration, or reentry settings. One thing that stands out in the research is that these behaviors are not only explained as random actions or just "bad choices." Researchers have measured attention bias, decision-making, and biological regulation in offender populations in more specific ways, and the findings suggest that these processes can be studied and in some cases modified. This matters for my proposal because I am interested in justice-impacted populations, but I also want the study to stay practical and usable in a real community setting.

The first article I looked at was by Zhao et al. (2022), which studied whether Attention Bias Modification, or ABM, could reduce aggression in young offenders with antisocial tendencies. The researchers had 84 institutionalized male young offenders between the ages of 16 and 18, and they randomly assigned them to an ABM group, a placebo group, or a waiting-list group. Before and after the treatment, the researchers measured aggressive behavior, attention bias toward positive stimuli, and attention bias toward hostile stimuli. They used measures such as the antisocial personality disorder subscale of the PDQ-4, the Aggression Questionnaire, an emotional dot-probe task, and a hostile emotional Stroop task (Zhao et al., 2022). Their results showed that ABM was most helpful for young offenders who had higher hostile attention bias or lower positive attention bias at the start of the study. This article is useful for my topic because it gives a clear intervention model and directly connects hostile attention bias with aggression in a justice-involved sample.

The second article was by Tully et al. (2024), and it looked at violent offenders from more of a biological perspective. The researchers studied 30 violent male offenders with antisocial personality disorder, with and without psychopathy, and compared them with 21 healthy non-offenders. The main variable was the striatal glutamate:GABA ratio, which was measured using proton magnetic resonance spectroscopy. Tully et al. (2024) found that offenders with antisocial personality disorder had a significantly reduced striatal glutamate:GABA ratio compared with non-offenders. This article is not an intervention study like Zhao et al. (2022), but it is still important because it supports the idea that violent behavior and poor behavioral control are connected to measurable regulation problems, not just personality labels or moral judgment. The third article was by Kuhn et al. (2024), which studied risk-taking and decision-making in violent offenders. Their study included 22 male violent offenders and 24 male non-violent controls. Participants completed active and sham transcranial direct current stimulation, or tDCS, over the right dorsolateral prefrontal cortex and then completed the Balloon Analogue Risk Task during fMRI scanning. Kuhn et al. (2024) found that violent offenders showed less optimal decision-making than non-violent controls, and active tDCS improved decision-making and increased prefrontal activity in the offender group. This article is more advanced than what I would actually propose for a community college research project, but it is still relevant because it shows that decision-making problems in offender populations can be measured and can shift after an intervention.

Together, these articles suggest that aggression and poor behavioral control in justice-involved populations are measurable and possibly changeable. Zhao et al. (2022) provides the clearest model for my own proposed study because it directly uses ABM with young offenders. Tully et al. (2024) and Kuhn et al. (2024) give background support showing that offender aggression and decision-making problems are connected to larger self-regulation processes. I am not proposing brain imaging or tDCS, because that would not be realistic for the type of study I am designing, but those articles still help explain why a lower-cost cognitive intervention is worth testing.

For my proposed study, I hypothesize that justice-impacted young adults who complete Attention Bias Modification training will show lower hostile attention bias scores and lower aggression scores at posttest than justice-impacted young adults who complete a neutral attention-training control condition. I also expect that the ABM group will show slightly better behavioral control on a brief decision-making task, although aggression and hostile attention bias would be the main outcomes of the study.

Methods

Participants

I would recruit approximately 48 justice-impacted young adults between the ages of 18 and 25. For this study, justice-impacted would mean individuals who have previously been incarcerated in juvenile detention, jail, or prison and are currently living in the community during reentry. Participants would be recruited through community reentry organizations, peer support groups, and community college programs that serve formerly incarcerated students. To be included, participants would need to be at least 18 years old, able to read English, and currently out of custody. Participants would be excluded if they reported an active psychotic disorder, major cognitive impairment, or acute intoxication at the time of testing, because those issues could interfere with task completion and informed consent. Participation would be completely voluntary.

Design

This study would use a between-groups pretest-posttest experimental design. Participants would be randomly assigned to either an Attention Bias Modification group or a neutral attention-training control group. The main independent variable would be training condition. The dependent variables would be hostile attention bias and aggressive behavior, measured before and after the intervention. A short computerized decision-making task would also be included as a secondary exploratory outcome because the offender decision-making literature suggests this is another area where change may matter. Zhao et al. (2022) used a randomized intervention design with young offenders, and that structure works well as the direct model for this proposal.

Materials

Participants would complete a short demographic questionnaire asking about age, gender, race or ethnicity, time since release, type of correctional setting, and whether they are currently enrolled in school or working. The main attention measure would be a brief hostile emotional Stroop task modeled after Zhao et al. (2022). In this task, participants would view hostile and neutral words presented in different colors and would identify the color of each word as quickly as possible. A hostile attention bias score would be created by subtracting reaction times for neutral words from reaction times for hostile words. Higher positive scores would indicate greater interference from hostile material, which would reflect greater hostile attention bias.

Aggressive behavior would be measured with an aggression questionnaire based on the domains used in the offender literature summarized by Zhao et al. (2022) and Kuhn et al. (2024). The questionnaire would assess physical aggression, verbal aggression, anger, and hostility. Higher total scores would indicate higher aggression. Zhao et al. reported strong internal consistency for their aggression measure in a young offender sample, which makes this kind of measure appropriate for a proposed study like this.

As a secondary measure, participants would complete a short computerized decision-making task based on the type of risk and reward task used by Kuhn et al. (2024). I would keep this part simple and use it only as an exploratory outcome, not as the main dependent variable. The main reason for including it is that Kuhn et al. found that violent offenders showed less optimal decision-making than controls, and that behavioral control could be measured after an intervention.

Procedure

After recruitment, eligible participants would attend an in-person pretest session. They would first review and sign an informed consent form. After consent, participants would complete the demographic questionnaire, the hostile emotional Stroop task, the aggression questionnaire, and the brief decision-making task. They would then be randomly assigned to either the ABM group or the control group. The ABM group would complete a visual search training program designed to direct attention toward positive targets and away from hostile distractors. The control group would complete a neutral visual search task that would be similar in structure and time, but would not train attention away from hostile content.

Participants would complete four total training sessions across two weeks. Each session would last about 15 to 20 minutes. After the final training session, participants would complete the same posttest measures used at baseline. Pretest and posttest scores would be compared across groups to determine whether the ABM condition produced greater reductions in hostile attention bias and aggression than the control condition. After finishing the study, participants would be debriefed and thanked, and they would receive a small gift card for participating.

Because this study would involve formerly incarcerated young adults, the consent process would need to be especially clear. Participation would not be connected in any way to probation, parole, court status, housing, treatment access, or school standing. Recruitment would not happen through correctional officers or anyone with direct authority over participants, because that would create pressure and would weaken voluntary consent. Participants would be told that they could stop the study at any time without any penalty.

Confidentiality would be protected by assigning each participant an ID number and storing names separately from study data. Results would only be reported at the group level. The study would not use deception. At the end of participation, each person would receive a short debriefing form and a list of counseling and reentry support resources in case the study topics brought up discomfort.

Discussion

If I were to carry out this study, I would expect the Attention Bias Modification group to show a greater reduction in hostile attention bias and aggression from pretest to posttest than the control group. I would also expect the ABM group to show somewhat better behavioral control on the decision-making task, although this would be secondary to the main aggression and attention bias results.

If the results supported the hypothesis, it would suggest that attention processes may be a practical place to intervene in reentry settings. This matters because most reentry programs do not have access to expensive biological treatments, but they may have access to basic computerized training and quiet space. One limitation would be that aggression would still be measured partly through self-report, and the intervention period would be short. Another limitation is that a community reentry sample would not be identical to the institutionalized youth sample studied by Zhao et al. (2022), so the results would need to be interpreted carefully.

References

Kuhn, L., Choy, O., Keller, L., Habel, U., & Wagels, L. (2024). Prefrontal tDCS modulates risk-taking in male violent offenders. *Scientific Reports, 14, Article 10087*. https://doi.org/10.1038/s41598-024-60795-z

Tully, J., Pereira, A. C., Sethi, A., Griem, J., Cross, B., Williams, S. C. R., Blair, R. J., Murphy, D., & Blackwood, N. (2024). Impaired striatal glutamate/GABA regulation in violent offenders with antisocial personality disorder and psychopathy. *Molecular Psychiatry, 29, 1824-1832*. https://doi.org/10.1038/s41380-024-02437-4

Zhao, Z., Yu, X., Ren, Z., Zhang, L., & Li, X. (2022). The remediating effect of Attention Bias Modification on aggression in young offenders with antisocial tendency: A randomized controlled trial. *Journal of Behavior Therapy and Experimental Psychiatry, 75, Article 101711*. https://doi.org/10.1016/j.jbtep.2021.101711